Next-Day Wildfire Spread Prediction via Tabular Machine Learning and Spatial Deep Learning: A Leakage-Aware Comparison on the NDWS Benchmark

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Abstract—We predict, for each 1 km pixel of an area around an active fire, whether it burns the next day, using the Next Day Wildfire Spread (NDWS) dataset. We compare baselines and four tabular learners on 32 engineered features with a compact CNN, a tuned U-Net and two kinds of hybrid. Models are tested on the official split, a location-disjoint split and a strict split disjoint in both location and week, because about 95% of official test patches share terrain with training data. XGBoost reaches test PR-AUC 0.265 on the official split and 0.141 on the spatial split, and LightGBM reaches 0.124 on the strict split. No network or hybrid exceeds the best tree on any split. The hybrid selected on validation data scores 0.259, 0.140 and 0.115; it falls significantly below XGBoost on the official split and is statistically indistinguishable from XGBoost on the other two. The strict split contains only 405 test patches, so its uncertainty is large. Training a U-Net on full patches raises PR-AUC by +0.084 to +0.103, which we attribute to the patch-centre position leak and which we therefore report only as a diagnostic. Tests of the tabular models suggest that they rely little on this leak. Removing the day-t fire channels reduces U-Net validation PR-AUC from 0.239 to 0.030, making it the dominant input. These results show that careful feature engineering remains competitive with spatial networks on this benchmark and that the choice of split changes the conclusions that can be drawn.

Index Terms—wildfire spread, gradient boosting, U-Net, remote sensing, spatial leakage, imbalanced segmentation, stacking

I. Introduction

Wildfire agencies need to know where a fire may be tomorrow to place crews, issue evacuations and plan containment lines. Physics-based simulators such as FARSITE [1], built on Rothermel's spread model [2], need detailed fuel and wind inputs and are expensive to run as ensembles. The Next Day Wildfire Spread (NDWS) dataset [3] instead frames next-day spread as per-pixel prediction from remote-sensing, weather and terrain rasters.

Tree ensembles on engineered features and convolutional networks approach this problem differently. Trees see one pixel at a time and depend on hand-made summaries of its surroundings. Networks can learn spatial patterns directly. This project compares the two on identical pixels, tests whether combining them helps, and asks how much of the measured skill survives when test data is genuinely separated from training data.

Our contributions are:

1) A 32-feature tabular representation in which every feature uses day-t inputs only, compared across a Dummy baseline, persistence baselines, logistic regression, random forest, XGBoost and LightGBM.
2) A terrain-fingerprint audit showing that about 95% of official test patches share terrain with training patches, with a location-disjoint split and a stricter split disjoint in both location and week.
3) A test of the patch-centre position leak, applied to both the tabular models and the networks.
4) A compact dilated CNN and a tuned U-Net, feature-assisted and late-fusion hybrids, and a unified comparison with patch-level bootstrap intervals, paired tests, three training seeds and ablations.

II. Problem Statement

For each 1 km pixel of a 64×64 patch containing an active fire on day t, estimate the probability that the pixel is burning on day $t+1$, using only information available at the end of day t. Two targets are reported. Active is the day-$t+1$ fire mask on every pixel with a known label, which includes fires that persist. Transition is the same mask restricted to pixels known not to be burning on day t, which isolates new fire. About 1.10% of training pixels are positive, so the task is strongly imbalanced. The models estimate statistical association and should not be read as evidence of how a fire physically propagated.

III. Research Objectives

1) Establish strong tabular baselines using gradient boosting on engineered features.
2) Develop spatial deep models that use the full raster context.

3) Measure whether either family's advantage holds on data separated by location and week.
4) Measure how much performance depends on the patch-centre position leak.
5) Test whether a hybrid of the two families improves on both, and report the outcome either way.

IV. Literature Review

Operational spread models descend from Rothermel's semi-empirical equation [2], and simulators such as FARSITE [1] propagate perimeters over gridded fuel and weather. Jain et al. [4] survey machine learning in wildfire science. Early deep models [5], [6] showed that CNNs can learn spread from rasters, and Sayad et al. [7] showed that standard classifiers can predict fire occurrence from remote-sensing features.

Huot et al. [3] released NDWS and reported test PR-AUC values of 11.5% for persistence, 19.8% for logistic regression on a pixel and its neighbours, 22.5% for random forest and 28.4% for a convolutional autoencoder. WildfireSpreadTS [8] extends the task to multi-day time series. Gradient boosting [9], with implementations such as XGBoost [10] and LightGBM [11], and random forests [12] remain strong on tabular problems. Grinsztajn et al. [13] show that tree ensembles still beat deep networks on typical medium-sized tabular data.

Fully convolutional networks [14] and the U-Net [15] produce per-pixel output, and dilated convolutions [16] widen the receptive field without pooling. Dice loss [17] targets region overlap under imbalance. Kayhan and van Gemert [18] show that boundary effects let convolutions exploit absolute position, and Liu et al. [19] study how CNNs handle coordinates. Stacked generalisation [20] learns how to combine base models on data they were not fitted on.

For evaluation, Saito and Rehmsmeier [21] show that precision-recall analysis is more informative than ROC under imbalance, and Roberts et al. [22] show that random cross-validation inflates estimates on spatially autocorrelated data. We know of no published NDWS evaluation that tests whether the official split is spatially independent or that measures the patch-centre leak in tabular models.

V. Dataset Description

NDWS contains 18,545 patches of 64×64 pixels at 1 km, covering the contiguous US from 2012 to 2020 [3]. Each patch has 11 day-t environmental rasters: SRTM elevation; GRIDMET wind direction and speed, minimum and maximum temperature, specific humidity and precipitation [23]; drought index; VIIRS NDVI; population density; and energy release component. It also has the day-t MODIS fire mask, and the target is the day-$t+1$ MODIS mask. The Kaggle release contains no coordinates, dates or event identifiers.

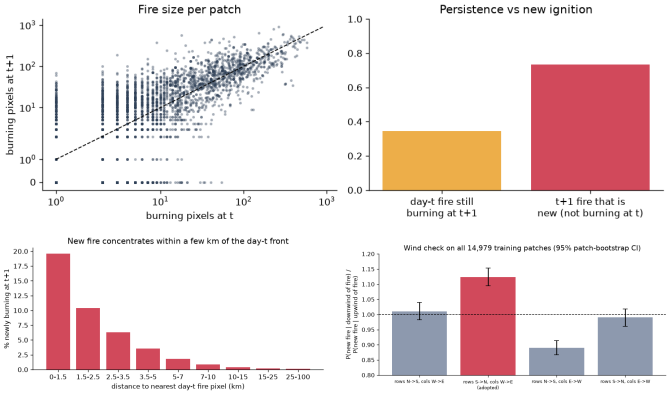


Fig. 1. Fire dynamics in NDWS: fire persistence and turnover (top), probability of next-day fire by distance to current fire (left), and the wind-orientation test (right).

Masks use 1 for fire, 0 for no fire and -1 for unknown. About 2.3% of target pixels are unknown and are excluded from training and evaluation. The official split assigns random weeks to train, validation and test (8:1:1) with a one-day buffer, giving 14,979, 1,877 and 1,689 patches.

VI. Data Preprocessing

Channels are cleaned identically for both model families. Temperatures outside 200–330 K, non-positive humidity and negative wind speeds become missing. Wind directions outside $[0, 361]$ degrees become missing, and values in $(360, 361]$ are wrapped. Precipitation is clipped at 0, NDVI is rescaled, and population is log-transformed. Trees receive missing values natively, and logistic regression and random forest use median imputation fitted on training data.

For the networks, channels are standardised using training-split statistics only, and missing values become 0 after standardisation, which is the training mean. Wind is expressed as an east/north vector and aspect as sine and cosine components, so geometric augmentation can transform them consistently. Vector channels are scaled but not mean-centred so that flipping an image negates the physical vector.

Since Kaggle provides no georeferencing, orientation was inferred from the data. Only with rows running south to north is new fire significantly more common downwind than upwind (ratio 1.12, 95% CI 1.09–1.15).

VII. Exploratory Data Analysis

About 74% of next-day fire pixels were not burning on day t, so persistence recalls only about 26% of them. New fire is concentrated near existing fire and becomes rare beyond a few kilometres (Fig. 1). The position of a pixel within a patch also relates to its label, because patches are centred on fire.

TABLE I
Tabular features by group

Group	Count	Features
Raw	10	elevation, wind speed, temperature min/max, humidity, precipitation, drought, NDVI, log population, ERC
Fire	8	day-t fire and unknown flags, fire share in windows of 3, 5, 9 and 17, distance to nearest fire, log fire count
Wind	6	east/north wind, fire up/downwind within 3 km, fire–wind alignment, wind push
Terrain	4	slope, aspect (sin, cos), upslope from fire
Weather	2	temperature range, vapour-pressure deficit
Vegetation	2	5×5 NDVI mean and standard deviation

VIII. Feature Engineering

Tabular models receive 32 features per pixel in six groups (Table I). Every feature is computed from day-t inputs of the same patch. Wind alignment is the cosine between the wind vector and the vector from the nearest fire to the pixel, and wind push scales that value by speed and decays with distance. Vapour-pressure deficit follows the standard FAO formulation [24].

IX. AML Methodology

Baselines are Dummy (the training positive rate), persistence and neighbourhood persistence. Learners are logistic regression, random forest, XGBoost and LightGBM. Training keeps every positive pixel and 2% of negatives. Sampling negatives at rate r multiplies the odds by $1/r$, so scores from unweighted models are corrected to the true prior [25]:

$$p = \frac{r\,p_s}{r\,p_s + 1 - p_s}. \tag{1}$$

Logistic regression and random forest choose between this correction and importance weighting on validation data. Boosting uses a learning rate of 0.05, up to 3000 trees and early stopping after 100 rounds without improvement on validation PR-AUC. A small validation grid selects tree size and leaf regularisation per split. Validation and test keep every known pixel.

X. Deep Learning Architecture

Both networks are fully convolutional, so they can train on crops and predict on full patches. The compact CNN stacks three two-layer convolution blocks with dilation 1, 2 and 4, giving a 29-pixel receptive field and 81,097 parameters. The U-Net uses three pooling stages with 24, 48, 96 and 192 channels, transposed convolutions and skip connections, with 1,086,841 parameters. Convolutions use batch normalisation [26], and the bottleneck uses spatial dropout [27]. Base models receive the 13 day-t channels.

The loss is applied to known pixels only. For a pixel with label y and predicted probability p, weighted binary cross-entropy is

$$\ell = -\left[w\,y \log p + (1-y)\log(1-p)\right], \tag{2}$$

TABLE II
Evaluation protocols: whether weeks and terrain are kept apart

Protocol	Weeks	Terrain	Train / val / test
Official	Yes	No	14,979 / 1,877 / 1,689
Spatial	No	Yes	14,822 / 1,855 / 1,868
Strict	Yes	Yes	12,050 / 389 / 405

and soft Dice loss is $1 - 2\sum py/(\sum p + \sum y)$.

Training uses AdamW [28] with learning rate 10^{-3}, batches of 64, a plateau scheduler, and up to 30 epochs with early stopping after 6 epochs without improvement in validation PR-AUC. Each batch receives one of eight flips and transposes, with vector channels transformed consistently. To prevent position leakage, training uses a random 32×32 crop of each patch.

XI. Hybrid Methodology

Three forms of hybrid are considered. Spatial context for tabular models is the existing feature set and is measured by the tabular ablations. The feature-assisted U-Net adds the engineered channels to the network input, for 31 channels and 1,090,729 parameters. Late fusion combines a tree and a network with logistic regression on logits:

$$\operatorname{logit} p = a + b_1 \operatorname{logit} p_{\text{tree}} + b_2 \operatorname{logit} p_{\text{net}}. \tag{3}$$

Following stacked generalisation [20], weights are fitted on half of the validation patches, so base models are combined using data they were not trained on. The remaining validation patches choose the hybrid among the fusion candidates and the feature-assisted U-Net, and set its threshold. Test data is used only for the final score. The tree input is the protocol's best tree on validation.

XII. Experimental Setup

A. Evaluation protocols

Elevation barely changes between days, so patches with identical non-flat 8×8 km elevation blocks were linked into 845 location groups. A mirrored-patch control produced no matches. About 95% of official test patches share terrain with training patches. We report three protocols (Table II). A separate dev protocol (12,050 training and 1,466 validation patches, all from official training weeks) tunes the networks, so no reported test patch influences their settings. The two largest location groups hold 71% of patches and always remain in training, so location-disjoint test sets contain smaller and more isolated fires.

B. Position-leak controls

Patches are centred on fires, so position relates to the label, and a network given full patches can infer its position from the padded border [18]. This signal would not exist in a real forecast. Three controls are used. Networks train on random crops. A position-controlled regime scores every model on one fixed random 32×32 window per held-out patch, with features recomputed from

TABLE III
Test PR-AUC at Phase 1 and in the final re-run

Model	Official (P1)	Official (final)	Spatial (P1)	Spatial (final)	Strict (new)
Persistence	0.107	0.107	0.037	0.037	0.027
Logistic regression	0.230	0.230	0.117	0.117	0.095
Random forest	0.265	0.265	0.135	0.135	0.120
XGBoost	0.265	0.265	0.141	0.141	0.115
LightGBM	0.263	0.263	0.138	0.138	0.124

the window alone. A patch-centre prior and a full-patch U-Net measure the leak directly. For tabular models we also add position columns, try to predict position from the features, and retrain on windows.

C. Tuning and repeated runs

Settings are chosen greedily, one factor at a time, on the dev protocol: architecture, loss, dropout, weight decay, width and feature channels. Architecture and channels are recorded without changing the selected base U-Net. Selected settings are then frozen. Final CNN, U-Net and feature-assisted U-Net runs use three seeds; hybrid inputs and thresholded results use the first seed. All experiments run from one script (scripts/run_all.sh) with versioned YAML configurations.

D. Changes since Phase 1

The graded Phase 1 materials are preserved unchanged in phase1_submission/. Table III compares their test PR-AUC with a full re-run. The official and spatial protocols reproduce to three decimals. Three changes were made after Phase 1. A strict protocol was added because the Phase 1 spatial split still mixed weeks between training and test. A position-leak test was added because Phase 1 applied crop training only to its U-Net and did not test whether tabular models exploit position. A Dummy baseline, a CNN, a tuned U-Net and hybrids were added to complete the comparison.

XIII. Evaluation Metrics

The primary metric is PR-AUC, computed on every known pixel without subsampling. A constant predictor scores the positive rate, so we also report lift (PR-AUC divided by the positive rate), which allows comparison across test sets with different rates. We also report ROC-AUC, precision, recall, F1, recall at 1% and 5% false-positive rates, confusion matrices and Brier score. Thresholds come from validation data. Pixels within a patch are correlated, so 95% intervals come from a patch-level bootstrap of 500 resamples. Differences between models use a paired bootstrap on identical resamples.

XIV. Results

A. Main comparison

Tables IV–VI give test results for the active target on full patches, and Fig. 2 summarises PR-AUC. Network rows and hybrid inputs use the first training seed, and Section XIV-F reports variation across seeds. Lift differs between protocols more modestly than raw PR-AUC because positive rates differ (1.25%, 0.48% and 0.65%). No network or hybrid exceeds the best tree on any split.

Table VII tests differences against the tree selected on validation, which is XGBoost on all three protocols. U-Net scores below XGBoost by -0.008 on the official split and by -0.011 on the spatial split, and in both cases the intervals exclude zero. On the strict split U-Net is also below XGBoost by -0.014. LightGBM has the highest strict test score, although the validation-selected tree is XGBoost, so the tests on that split use the validation-selected model and not the best test score.

B. Position-controlled scoring

Table VIII scores each model on fixed random windows, with features recomputed from the window. Here the position prior has lift of 1.1, 1.1 and 1.1, which confirms that position carries almost no information within these windows. On the official protocol the hybrid exceeds the window-trained LightGBM by +0.012 (95% CI +0.006 to +0.017), the only comparison here in which a hybrid significantly beats a tree. On the spatial protocol the models are close. On the strict protocol U-Net is below window-trained LightGBM by -0.015 and the hybrid by -0.009, and both intervals exclude zero (Table IX).

C. New fire only

Table X reports PR-AUC for pixels not burning on day t. Persistence is equal to the positive rate by construction. On the official split U-Net exceeds XGBoost by +0.005 (+0.000 to +0.009). On the spatial and strict splits XGBoost is ahead, by -0.012 and -0.012 for U-Net, so the small network advantage for new fire does not carry over to held-out locations.

D. Patch-centre leak

A position-only patch-centre prior reaches PR-AUC 0.135, 0.121 and 0.137 on full patches. On the strict split it is higher than every valid model, which shows how strongly full-patch scores depend on patch centring. Training the U-Net on full patches raises its PR-AUC to 0.341, 0.232 and 0.190, gains of +0.084, +0.103 and +0.089. These runs are diagnostics and are not valid forecasters.

For tabular models (Table XI), adding explicit position raises PR-AUC substantially, so a usable position signal exists that the features do not already contain. A separate model can recover closeness to the centre from the features with ROC-AUC 0.843, 0.742 and 0.753, so the features do partly encode it, most likely through the location of the fire itself. On position-controlled windows the full-patch model and the window-trained model score similarly, which suggests that tabular results are not driven by position.

E. Hybrids

Table XII shows every candidate. Selection on the held-out half of validation data chose Fusion: tree + feature U-Net on the official and spatial protocols and Fusion:

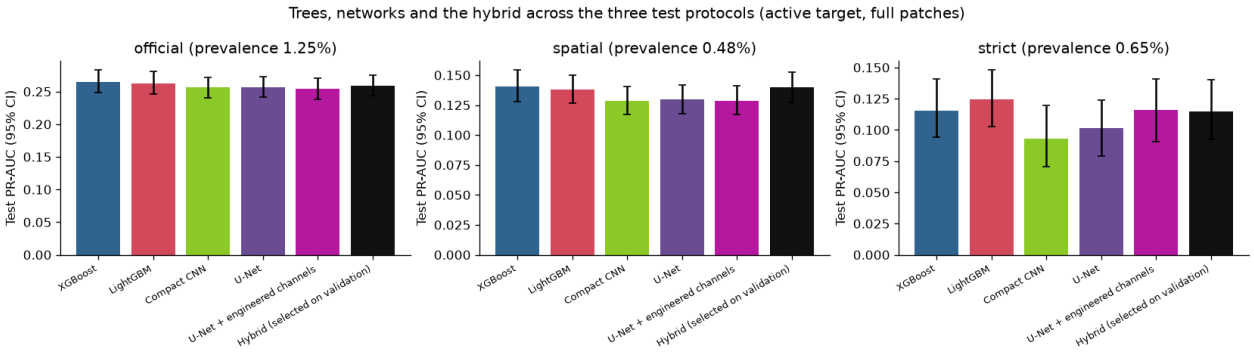


Fig. 2. Test PR-AUC with 95% patch-bootstrap intervals for trees, networks and the validation-selected hybrid.

TABLE IV
Official protocol, active target, full patches. Thresholded metrics use validation thresholds.

Model	PR-AUC [95% CI]	Lift	ROC-AUC	Prec.	Rec.	F1	R@1%	R@5%
Dummy (training rate)	0.013 [0.012, 0.014]	1.0	0.500	0.013	1.000	0.025	0.010	0.050
Persistence	0.107 [0.097, 0.118]	8.5	0.633	0.357	0.273	0.309	0.276	0.305
Logistic regression	0.230 [0.213, 0.249]	18.4	0.903	0.246	0.487	0.327	0.363	0.665
Random forest	0.265 [0.248, 0.282]	21.1	0.901	0.261	0.498	0.343	0.389	0.679
XGBoost	0.265 [0.248, 0.283]	21.1	0.911	0.242	0.528	0.332	0.389	0.680
LightGBM	0.263 [0.246, 0.281]	21.0	0.905	0.245	0.525	0.334	0.387	0.680
Compact CNN	0.256 [0.240, 0.272]	20.4	0.897	0.242	0.529	0.332	0.385	0.675
U-Net	0.257 [0.242, 0.272]	20.5	0.892	0.275	0.486	0.351	0.392	0.677
U-Net + engineered channels	0.254 [0.239, 0.270]	20.3	0.915	0.258	0.500	0.340	0.380	0.677
Hybrid (selected on validation)	0.259 [0.244, 0.276]	20.7	0.917	0.254	0.514	0.340	0.385	0.682

TABLE V
Spatial protocol, active target, full patches.

Model	PR-AUC [95% CI]	Lift	ROC-AUC	Prec.	Rec.	F1	R@1%	R@5%
Dummy (training rate)	0.005 [0.005, 0.005]	1.0	0.500	0.005	1.000	0.010	0.010	0.050
Persistence	0.037 [0.032, 0.041]	7.6	0.572	0.224	0.146	0.177	0.152	0.186
Logistic regression	0.117 [0.106, 0.127]	24.2	0.867	0.193	0.272	0.226	0.374	0.639
Random forest	0.135 [0.123, 0.149]	28.1	0.885	0.191	0.304	0.234	0.382	0.627
XGBoost	0.141 [0.128, 0.154]	29.1	0.899	0.230	0.263	0.246	0.382	0.628
LightGBM	0.138 [0.127, 0.150]	28.6	0.896	0.229	0.261	0.244	0.382	0.621
Compact CNN	0.129 [0.117, 0.141]	26.6	0.846	0.193	0.293	0.233	0.366	0.556
U-Net	0.130 [0.118, 0.142]	26.9	0.841	0.192	0.299	0.234	0.370	0.580
U-Net + engineered channels	0.129 [0.117, 0.141]	26.6	0.873	0.196	0.277	0.230	0.345	0.571
Hybrid (selected on validation)	0.140 [0.127, 0.152]	29.0	0.893	0.220	0.273	0.244	0.380	0.607

tree + U-Net on the strict protocol. The selected hybrid is below XGBoost by -0.006 on the official split, where the interval excludes zero, and within -0.001 and -0.001 on the spatial and strict splits, where the intervals contain zero. It exceeds U-Net on the spatial and strict splits. Some candidates not selected had higher test scores, but choosing them on that basis would use test data. The hybrid therefore improves on the U-Net but not on the trees.

F. Variation across training seeds

Table XIII shows test PR-AUC across three seeds. Seed-to-seed standard deviation is 0.001 to 0.013, of a similar size to several differences between models. Feature-assisted U-Net has means of 0.254, 0.135 and 0.116 against U-Net's 0.252, 0.128 and 0.104. Its advantage on the strict split is the largest difference but also has the largest spread, so we consider it suggestive rather than established.

G. Prediction maps and errors

Fig. 3 shows example strict-protocol maps. XGBoost and the hybrid concentrate probability close to existing fire, and networks produce wider, smoother regions. Missed fire is mostly new fire away from the day-t perimeter. At the validation-selected threshold, XGBoost recalls 0.95, 0.72 and 0.66 of fire within 1.5 km of a current fire pixel on the official, spatial and strict splits, then falls to 0.45, 0.19 and 0.06 between 1.5 and 3.5 km and to 0.01, 0.01 and 0.00 between 3.5 and 7 km. Confusion matrices and error by distance are shown in Fig. 4.

TABLE VI
Strict protocol, active target, full patches.

Model	PR-AUC [95% CI]	Lift	ROC-AUC	Prec.	Rec.	F1	R@1%	R@5%
Dummy (training rate)	0.006 [0.005, 0.007]	1.0	0.500	0.006	1.000	0.013	0.010	0.050
Persistence	0.027 [0.021, 0.035]	4.2	0.554	0.194	0.112	0.142	0.118	0.154
Logistic regression	0.095 [0.079, 0.113]	14.8	0.868	0.182	0.210	0.195	0.279	0.560
Random forest	0.120 [0.097, 0.145]	18.6	0.856	0.207	0.190	0.198	0.311	0.545
XGBoost	0.115 [0.094, 0.141]	17.8	0.866	0.222	0.185	0.202	0.320	0.554
LightGBM	0.124 [0.103, 0.148]	19.3	0.879	0.204	0.214	0.209	0.324	0.596
Compact CNN	0.093 [0.071, 0.119]	14.4	0.831	0.183	0.244	0.209	0.270	0.446
U-Net	0.101 [0.079, 0.124]	15.7	0.783	0.200	0.198	0.199	0.296	0.496
U-Net + engineered channels	0.116 [0.091, 0.141]	17.9	0.811	0.203	0.266	0.230	0.319	0.493
Hybrid (selected on validation)	0.115 [0.093, 0.140]	17.8	0.853	0.222	0.192	0.206	0.316	0.552

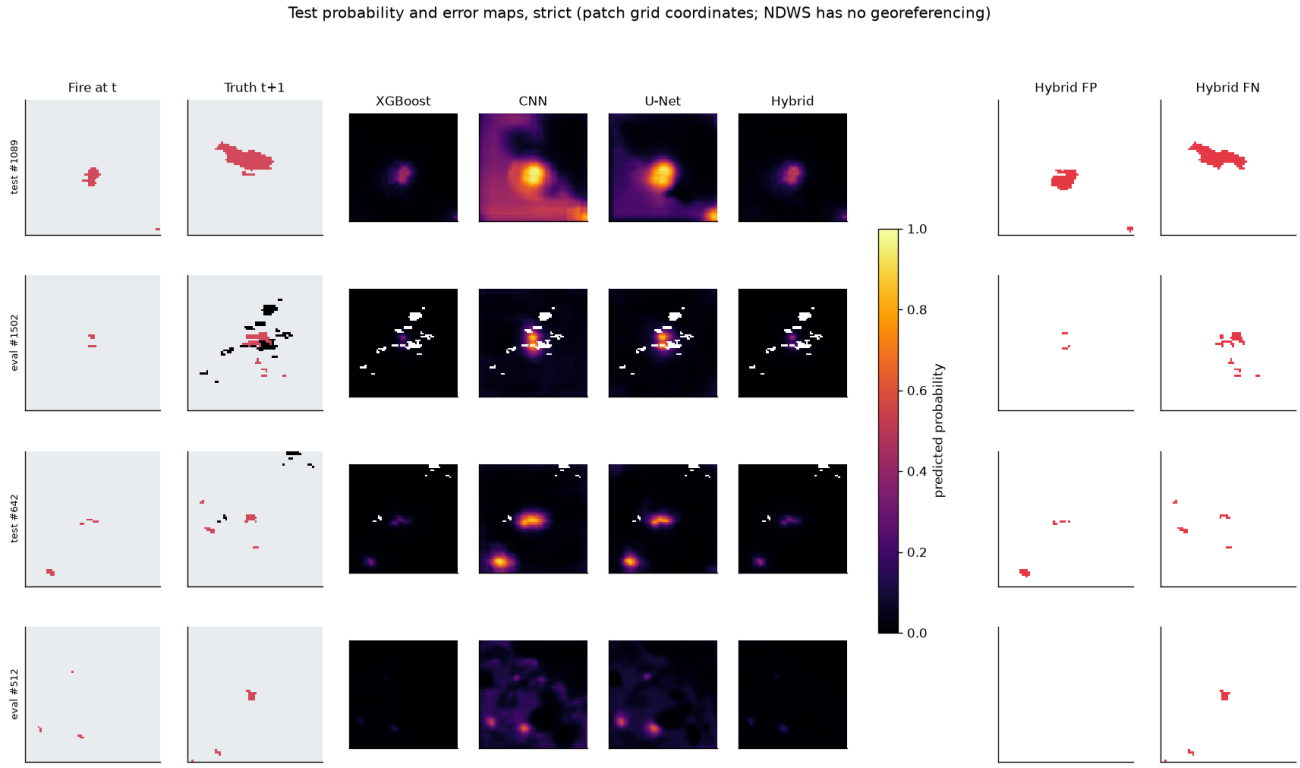


Fig. 3. Strict-protocol test maps: fire on day t, actual day $t+1$ fire, XGBoost, CNN, U-Net, hybrid, and hybrid false positives and false negatives. Axes are patch-grid coordinates because NDWS has no georeferencing.

XV. Ablation Studies

Tabular ablations (Table XIV) show that raw values alone are weak (0.010 on spatial and 0.012 on strict test), and fire context alone already reaches 0.111 and 0.095. Removing fire context from the full model lowers strict test PR-AUC from 0.124 to 0.110, while removing wind leaves it at 0.124.

U-Net ablations are in Table XV and Fig. 5. Without day-t fire channels, validation PR-AUC falls from 0.239 to 0.030. Removing skip connections gives 0.228, removing weather gives 0.227 and removing wind gives 0.233. Removing augmentation gives 0.242, slightly above the selected model, so we do not claim that augmentation improved performance. These are single runs on the dev protocol, so small differences should not be interpreted. Table XVI records the tuning sweep.

XVI. Discussion

Why trees stay ahead. The ablations show that next-day fire depends mainly on the day-t fire and its immediate surroundings, and the engineered features describe this directly. Spatial networks can learn those patterns but must do so from about 10^4 patches, which may be why they do not surpass the trees. This agrees with Grinsztajn et al. [13], although we did not test this explanation directly.

The effect of the split. Moving from the official split to location-disjoint splits halves PR-AUC, partly because positive rates fall. Rankings between trees and networks

TABLE VII
Paired bootstrap differences in test PR-AUC, active target, full patches (first training seed).

Protocol	Comparison	ΔPR-AUC	95% CI	$P(\Delta \leq 0)$
official	unet - xgboost	-0.0083	[-0.0143, -0.0023]	1.000
official	cnn - xgboost	-0.0088	[-0.0144, -0.0031]	0.998
official	unet_features - xgboost	-0.0108	[-0.0155, -0.0068]	1.000
official	hybrid - xgboost	-0.0057	[-0.0096, -0.0025]	1.000
official	hybrid - unet	+0.0027	[-0.0023, +0.0080]	0.186
official	unet - cnn	+0.0005	[-0.0040, +0.0048]	0.392
official	unet_features - unet	-0.0025	[-0.0077, +0.0025]	0.832
official	unet_fullpatch - unet	+0.0844	[+0.0760, +0.0931]	0.000
spatial	unet - xgboost	-0.0108	[-0.0152, -0.0072]	1.000
spatial	cnn - xgboost	-0.0121	[-0.0181, -0.0061]	1.000
spatial	unet_features - xgboost	-0.0120	[-0.0171, -0.0070]	1.000
spatial	hybrid - xgboost	-0.0008	[-0.0039, +0.0024]	0.724
spatial	hybrid - unet	+0.0100	[+0.0070, +0.0138]	0.000
spatial	unet - cnn	+0.0012	[-0.0027, +0.0049]	0.268
spatial	unet_features - unet	-0.0012	[-0.0052, +0.0033]	0.696
spatial	unet_fullpatch - unet	+0.1026	[+0.0937, +0.1122]	0.000
strict	unet - xgboost	-0.0141	[-0.0199, -0.0096]	1.000
strict	cnn - xgboost	-0.0222	[-0.0303, -0.0137]	1.000
strict	unet_features - xgboost	+0.0005	[-0.0056, +0.0056]	0.466
strict	hybrid - xgboost	-0.0005	[-0.0025, +0.0011]	0.738
strict	hybrid - unet	+0.0135	[+0.0104, +0.0176]	0.000
strict	unet - cnn	+0.0082	[+0.0008, +0.0142]	0.020
strict	unet_features - unet	+0.0146	[+0.0088, +0.0210]	0.000
strict	unet_fullpatch - unet	+0.0886	[+0.0724, +0.1045]	0.000

TABLE VIII
Position-controlled test results: active target on fixed 32×32 windows.

Official

Model	PR-AUC [95% CI]	Lift	ROC-AUC
Position prior in window (leak probe)	0.032 [0.030, 0.036]	1.1	0.549
LightGBM (full-patch model)	0.293 [0.274, 0.316]	10.3	0.876
LightGBM (window-trained)	0.290 [0.273, 0.310]	10.1	0.871
Compact CNN	0.287 [0.269, 0.307]	10.0	0.867
U-Net	0.297 [0.278, 0.316]	10.4	0.876
U-Net + engineered channels	0.284 [0.265, 0.304]	9.9	0.853
Hybrid (selected on validation)	0.301 [0.283, 0.321]	10.5	0.878

Spatial

Model	PR-AUC [95% CI]	Lift	ROC-AUC
Position prior in window (leak probe)	0.015 [0.013, 0.016]	1.1	0.537
LightGBM (full-patch model)	0.174 [0.159, 0.189]	12.9	0.877
LightGBM (window-trained)	0.169 [0.154, 0.184]	12.5	0.853
Compact CNN	0.161 [0.145, 0.176]	11.9	0.822
U-Net	0.167 [0.150, 0.182]	12.3	0.840
U-Net + engineered channels	0.167 [0.151, 0.183]	12.3	0.832
Hybrid (selected on validation)	0.172 [0.155, 0.187]	12.7	0.855

Strict

Model	PR-AUC [95% CI]	Lift	ROC-AUC
Position prior in window (leak probe)	0.020 [0.016, 0.024]	1.1	0.535
LightGBM (full-patch model)	0.157 [0.128, 0.188]	8.7	0.847
LightGBM (window-trained)	0.148 [0.116, 0.182]	8.2	0.797
Compact CNN	0.123 [0.094, 0.157]	6.8	0.792
U-Net	0.134 [0.104, 0.166]	7.4	0.770
U-Net + engineered channels	0.143 [0.112, 0.177]	8.0	0.778
Hybrid (selected on validation)	0.139 [0.109, 0.172]	7.7	0.781

stay similar, which suggests that terrain overlap does not explain the trees' lead. Position-controlled scoring narrows the gap on the official and spatial splits: on official the U-Net and the hybrid match or exceed the window-trained tree, and on spatial their differences from it have intervals containing zero. On the strict split the tree remains significantly ahead of both.

Position. Full-patch scores contain a large position effect, as the centre prior and full-patch U-Net show. Scores from different papers can only be compared when position handling is the same. Published NDWS scores are also not directly comparable with ours because training and evaluation details differ.

TABLE IX
Paired differences on position-controlled windows (first training seed).

Protocol	Comparison	ΔPR-AUC	95% CI	$P(\Delta \leq 0)$
official	unet - lightgbm_crop	+0.0068	[-0.0005, +0.0142]	0.046
official	unet_features - lightgbm_crop	-0.0060	[-0.0119, -0.0006]	0.986
official	hybrid - lightgbm_crop	+0.0116	[+0.0058, +0.0171]	0.000
official	hybrid - unet	+0.0048	[+0.0026, +0.0070]	0.000
spatial	unet - lightgbm_crop	-0.0027	[-0.0070, +0.0013]	0.864
spatial	unet_features - lightgbm_crop	-0.0022	[-0.0069, +0.0028]	0.816
spatial	hybrid - lightgbm_crop	+0.0020	[-0.0008, +0.0052]	0.108
spatial	hybrid - unet	+0.0047	[+0.0013, +0.0080]	0.002
strict	unet - lightgbm_crop	-0.0145	[-0.0234, -0.0080]	1.000
strict	unet_features - lightgbm_crop	-0.0049	[-0.0128, +0.0023]	0.912
strict	hybrid - lightgbm_crop	-0.0091	[-0.0160, -0.0039]	1.000
strict	hybrid - unet	+0.0054	[+0.0038, +0.0075]	0.000

TABLE X
PR-AUC on the transition target (new fire only), full patches.

Model	official	spatial	strict
Dummy (training rate)	0.009	0.004	0.006
Persistence	0.009	0.004	0.006
Logistic regression	0.123	0.078	0.075
Random forest	0.135	0.087	0.087
XGBoost	0.141	0.090	0.084
LightGBM	0.138	0.089	0.092
Compact CNN	0.143	0.073	0.062
U-Net	0.145	0.078	0.072
U-Net + engineered channels	0.138	0.075	0.081
Hybrid (selected on validation)	0.142	0.087	0.084

Hybrids. Late fusion improved on the U-Net on two of three splits but did not exceed the trees. In the fusion of a tree and the base U-Net, the tree received the larger weight on all three splits. The feature-assisted U-Net received the larger weight when it was the network input.

Practical reading. At the chosen threshold, models mostly recover fire next to the current perimeter and rarely detect spread beyond a few kilometres. Fire appearing several kilometres from the current perimeter remains largely undetected at these thresholds.

XVII. Limitations

NDWS has no coordinates, dates, event labels, fuel moisture or suppression information. Weeks are assigned at random, so this work tests generalisation to held-out weeks and locations and does not test forecasting into future seasons. Weather spans all of day t, so a forecast can only be issued after that day ends. MODIS daily detections are affected by cloud, smoke and overpass timing, which adds label noise. Terrain matching detects only exact 8×8 km overlaps, so adjacent fires can remain on both sides of a split. The strict test set contains only 405 patches, so its intervals are wide. Samples exist only where fire is present, so results do not represent monitoring of all land. Sweep and ablation results are single runs on the dev protocol. The position controls reduce the leak but cannot prove that no position information remains in the features.

TABLE XI
Position-leak tests of LightGBM on test data.

Test	official	spatial	strict
Features only, PR-AUC	0.263	0.138	0.124
Features + position, PR-AUC	0.345	0.278	0.251
Recover position from features, ROC-AUC	0.843	0.742	0.753
Window: full-patch model, PR-AUC	0.293	0.174	0.157
Window: window-trained model, PR-AUC	0.290	0.169	0.148
Window: position prior, PR-AUC	0.032	0.015	0.020
Window: positive rate	0.029	0.014	0.018

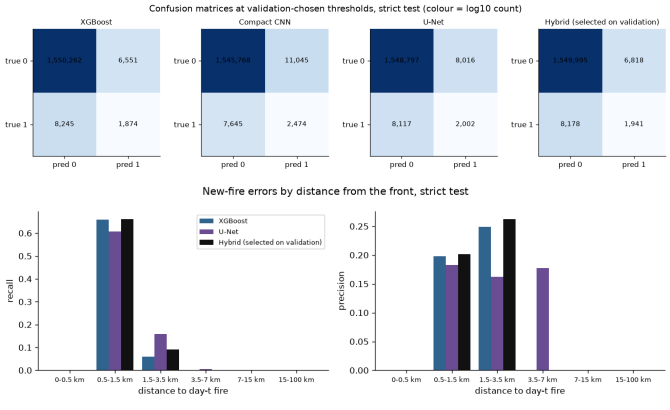


Fig. 4. Strict-protocol confusion matrices (top) and error by distance to the nearest day-t fire (bottom).

XVIII. Conclusion

Tree ensembles on engineered features were the strongest models on all three test protocols. A compact CNN, a tuned U-Net and hybrids came close but did not exceed them, and the differences for the networks are significant on the official and spatial splits. The hybrid selected without test data improved on the U-Net on two splits and matched XGBoost on the spatial and strict splits, but did not exceed the trees. The strict split and the position tests show that benchmark scores depend strongly on how data is separated and how patches are sampled. Each of the five research objectives was addressed, and the hybrid hypothesis was tested and not supported for improving on the best tree.

XIX. Future Work

Useful extensions include evaluation on WildfireSpreadTS [8], which provides dates and event identifiers and supports chronological splits; temporal models such as ConvLSTM [29] on multi-day inputs; fuel-moisture and fuel-type inputs; calibrated probability outputs for operational thresholds; and stronger location separation using true coordinates.

Reproducibility and Contributions

All code, configurations and result files are in the project repository. scripts/run_all.sh reproduces every experiment, and every number in this report is generated from the result files. Individual contributions are recorded in CONTRIBUTIONS.md.

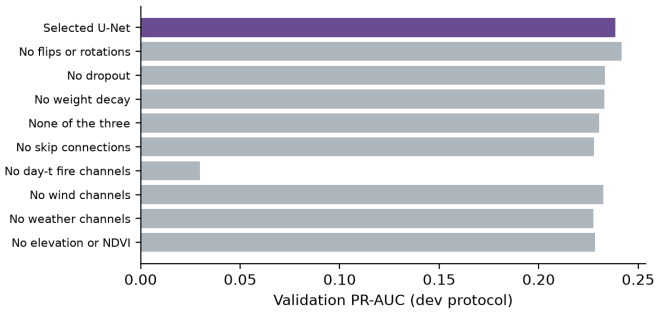


Fig. 5. Effect of each ablation on U-Net validation PR-AUC (dev protocol).

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TABLE XII
Hybrid candidates. Weights are fitted on half of validation data, the other half selects the hybrid, and test scores were not used for selection.

Protocol	Candidate	Weights on logits	Val. PR-AUC	Test PR-AUC
official	Fusion input: tree only	tree=0.91	0.192	0.265
official	Fusion input: U-Net only	unet=0.99	0.199	0.257
official	Fusion: tree + U-Net	tree=0.53, unet=0.43	0.205	0.270
official	Fusion: tree + CNN	tree=0.80, cnn=0.13	0.195	0.266
official	Fusion: tree + feature U-Net (selected)	tree=0.21, unet_features=0.81	0.205	0.259
official	U-Net + engineered channels	–	0.204	0.254
spatial	Fusion input: tree only	tree=1.02	0.228	0.141
spatial	Fusion input: U-Net only	unet=1.16	0.229	0.130
spatial	Fusion: tree + U-Net	tree=0.65, unet=0.42	0.236	0.142
spatial	Fusion: tree + CNN	tree=0.66, cnn=0.43	0.238	0.142
spatial	Fusion: tree + feature U-Net (selected)	tree=0.37, unet_features=0.85	0.242	0.140
spatial	U-Net + engineered channels	–	0.238	0.129
strict	Fusion input: tree only	tree=0.91	0.235	0.115
strict	Fusion input: U-Net only	unet=1.05	0.239	0.101
strict	Fusion: tree + U-Net (selected)	tree=0.68, unet=0.28	0.242	0.115
strict	Fusion: tree + CNN	tree=0.46, cnn=0.61	0.239	0.111
strict	Fusion: tree + feature U-Net	tree=0.42, unet_features=0.64	0.240	0.120
strict	U-Net + engineered channels	–	0.230	0.116

TABLE XIII
Test PR-AUC across three training seeds, active target, full patches.

Protocol	Model	Mean	SD	Min	Max
official	Compact CNN	0.255	0.006	0.248	0.260
official	U-Net	0.252	0.005	0.248	0.257
official	U-Net + engineered channels	0.254	0.002	0.251	0.256
spatial	Compact CNN	0.127	0.002	0.125	0.129
spatial	U-Net	0.128	0.001	0.127	0.130
spatial	U-Net + engineered channels	0.135	0.007	0.129	0.142
strict	Compact CNN	0.099	0.007	0.093	0.107
strict	U-Net	0.104	0.003	0.101	0.108
strict	U-Net + engineered channels	0.116	0.013	0.103	0.128

TABLE XIV
LightGBM feature-group ablations, PR-AUC.

Feature groups	Spatial val.	Spatial test	Strict val.	Strict test
Raw values only	0.025	0.010	0.038	0.012
Fire context only	0.191	0.111	0.186	0.095
Raw + fire	0.212	0.130	0.214	0.116
Raw + fire + wind	0.226	0.140	0.230	0.119
Raw + fire + wind + terrain	0.228	0.142	0.224	0.119
All features	0.230	0.138	0.232	0.124
All minus raw	0.226	0.138	0.234	0.116
All minus fire	0.213	0.131	0.216	0.110
All minus wind	0.220	0.139	0.217	0.124
All minus terrain	0.227	0.139	0.234	0.119
All minus weather	0.227	0.141	0.230	0.121
All minus vegetation	0.227	0.146	0.229	0.119

TABLE XV
U-Net ablations on the dev protocol, validation PR-AUC.

Change	Val. PR-AUC	Change vs selected
Selected U-Net	0.2386	–
No flips or rotations	0.2417	+0.0032
No dropout	0.2334	-0.0051
No weight decay	0.2330	-0.0056
None of the three	0.2305	-0.0081
No skip connections	0.2278	-0.0108
No day-t fire channels	0.0297	-0.2089
No wind channels	0.2325	-0.0060
No weather channels	0.2274	-0.0111
No elevation or NDVI	0.2284	-0.0102

TABLE XVI
Hyperparameter sweep on the dev protocol, validation PR-AUC.

Stage	Option	Val. PR-AUC	Best epoch	Best
architecture	unet	0.2356	13	✓
architecture	cnn	0.2329	17	
loss	bce	0.2356	13	
loss	weighted_bce (weight 3)	0.2268	14	
loss	weighted_bce (weight 10)	0.2386	6	✓
loss	dice	0.1728	5	
loss	bce_dice	0.2320	9	
dropout	0.0	0.2334	13	
dropout	0.2	0.2386	6	✓
dropout	0.4	0.2354	17	
weight_decay	0.0	0.2330	8	
weight_decay	0.0001	0.2386	6	✓
weight_decay	0.001	0.2281	6	
width	16	0.2339	8	
width	24	0.2386	6	✓
width	32	0.2279	13	
feature_channels	base	0.2386	6	
feature_channels	fire	0.2391	7	
feature_channels	fire_wind	0.2414	3	
feature_channels	all	0.2422	7	✓

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